BUSINESS INTELLIGENCE & TOOLS

EXPERIMENT – 3

DIMENSIONAL MODELING & STAR SCHEMA

Using Microsoft Power BI Desktop

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1. AIM

To design and implement a dimensional data model using Microsoft Power BI by transforming transactional data from the Global Superstore dataset into a Star Schema consisting of Fact and Dimension tables for efficient reporting and analytical processing.

2. OBJECTIVES

- To understand the concepts of Dimensional Modeling and Star Schema.
- To differentiate between Fact Tables and Dimension Tables.
- To transform transactional data into a structured dimensional model using Power BI.
- To create Fact and Dimension tables using Power Query Editor.
- To establish relationships between Fact and Dimension tables.
- To create DAX measures for analytical reporting.
- To validate the Star Schema using Power BI visualizations.

3. TOOLS / SOFTWARE REQUIRED

- Microsoft Power BI Desktop
- Global Superstore Dataset.xlsx
- Power Query Editor – built into Power BI Desktop
- DAX (Data Analysis Expressions)

4. DATASET USED

Global Superstore Dataset.xlsx

5. THEORY

5.1 Dimensional Modeling

Dimensional Modeling is a data warehouse design technique used in Business Intelligence systems to organize data for analytical processing and reporting. It separates measurable business events from descriptive information so that reports can be created efficiently and understood easily.

5.2 Fact Tables

Fact Tables contain measurable business events and numerical values such as Sales, Profit, Quantity and Discount. They also contain keys that connect each transaction to the relevant dimensions. In this experiment, Fact_Sales is the central fact table.

5.3 Dimension Tables

Dimension Tables contain descriptive attributes used to analyse facts. They provide context such as Customer, Product, Region and Date. The dimensions created in this experiment are Dim_Customer, Dim_Product, Dim_Region and Dim_Date.

5.4 Star Schema

A Star Schema is a dimensional model in which a central Fact Table is directly connected to multiple Dimension Tables. Fact_Sales is placed at the centre while the dimension tables surround it. This structure simplifies reporting, filtering and aggregation in Power BI.

5.5 DAX

DAX (Data Analysis Expressions) is the formula language used in Power BI for creating analytical measures. Measures such as Total Sales, Total Profit, Profit Margin and Average Order Value can respond dynamically to filters coming from the dimension tables.

5.6 Fact Table vs Dimension Table

Aspect	Fact Table	Dimension Table
Purpose	Stores measurable business events	Stores descriptive context
Typical data	Sales, Profit, Quantity, Discount	Customer, Product, Region, Date
Role	Provides values to aggregate	Provides categories for filtering/grouping
Example	Fact_Sales	Dim_Customer, Dim_Product, Dim_Region, Dim_Date

6. PROCEDURE

Part A: Load Dataset

Step 1: Open Microsoft Power BI Desktop.

Launch Power BI Desktop and create/open the Experiment 3 file.

Step 2: Import the Global Superstore dataset.

Home > Get Data > Excel Workbook > select Global Superstore Dataset.xlsx > connect/load the workbook.

Step 3: Select the required worksheets.

- Orders
- Returns
- People
- Verify that the imported tables are available in Power BI.

Part B: Identify Fact and Dimension Tables

Step 4: Analyse the Orders table.

Identify measurable attributes such as Sales, Profit, Quantity and Discount. These are suitable for the central Fact_Sales table.

Step 5: Identify descriptive attributes.

Identify Customer, Product, Region and Date attributes that will provide context for analysis.

Step 6: Categorize the data.

- Fact Table: Fact_Sales
- Dimension Tables: Dim_Customer, Dim_Product, Dim_Region and Dim_Date

Part C: Create Dimension Tables

Step 7: Open Power Query Editor.
Home > Transform Data.

Step 8: Create Dim_Customer.
Keep Customer ID, Customer Name and Segment. Remove duplicates so the customer key is unique.

Step 9: Create Dim_Product.
Keep Product ID, Product Name, Category and Sub-Category. Remove duplicate product records.

Step 10: Create Dim_Region.
Keep Region, Country, State and City. Remove duplicates and verify consistent location values.

Step 11: Create Dim_Date.
Create Order Date, Year, Quarter and Month and verify correct date formatting.Insert the Dim_Date screenshot.

Step 12: Remove duplicate records.
Verify that all dimension keys used in relationships contain unique values.

Part D: Create Fact Table

Step 13: Create Fact_Sales.
Create the central table containing Order ID, Customer ID, Product ID, Region, Order Date, Sales, Quantity, Profit and Discount.

Step 14: Verify key columns.
Check that all foreign-key and measure columns are present and correctly typed.

Part E: Build Star Schema

Step 15: Open Model View.
Select Model View from the left navigation pane.

Step 16: Create relationships.

- Fact_Sales ↔ Dim_Customer
- Fact_Sales ↔ Dim_Product
- Fact_Sales ↔ Dim_Region
- Fact_Sales ↔ Dim_Date
- Use Many-to-One relationships with Fact_Sales on the many side and each dimension on the one side.

Step 17: Verify relationship keys.
Confirm that every dimension key connects to the corresponding foreign-key field in Fact_Sales and that relationships are active.

Step 18: Arrange the model.
Place Fact_Sales in the centre and arrange the four dimensions around it to form a star-shaped model.

Part F: Validate Model and Create DAX Measures

Step 19: Create Power BI visualizations.
Create visuals such as Sales by Category, Profit by Region and Sales Trend by Year to validate the dimensional model.

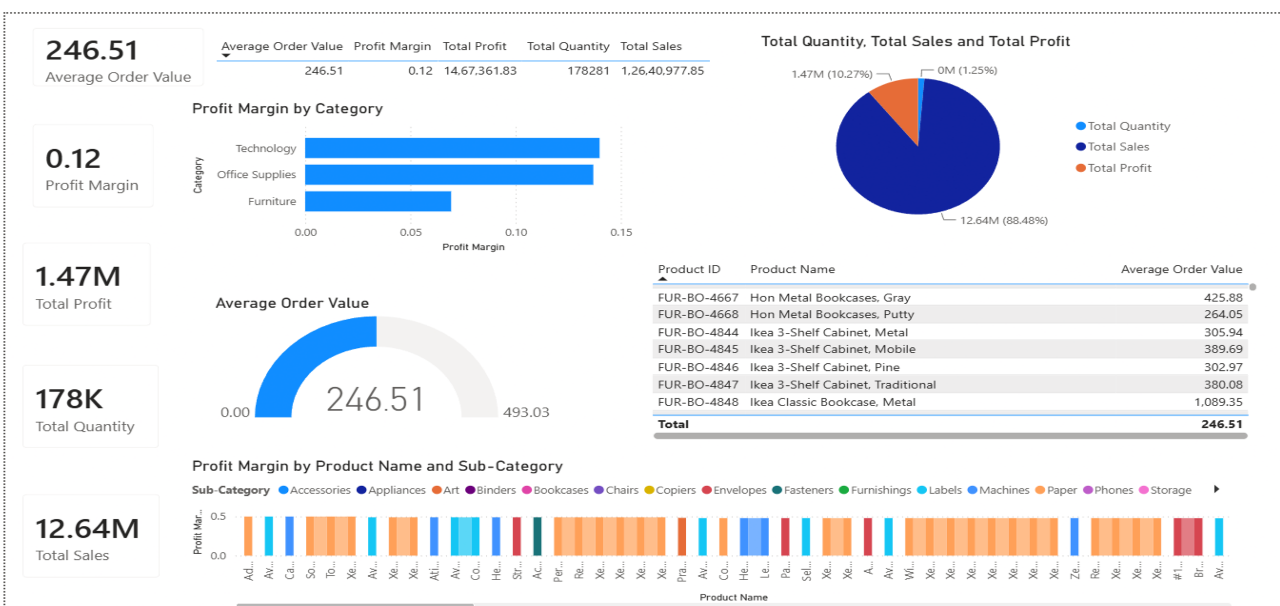
Step 20: Create DAX measures.

- Total Sales = SUM(Fact_Sales[Sales])
- Total Profit = SUM(Fact_Sales[Profit])
- Profit Margin = DIVIDE(SUM(Fact_Sales[Profit]), SUM(Fact_Sales[Sales]))
- Average Order Value = DIVIDE([Total Sales], DISTINCTCOUNT(Fact_Sales[Order ID]))

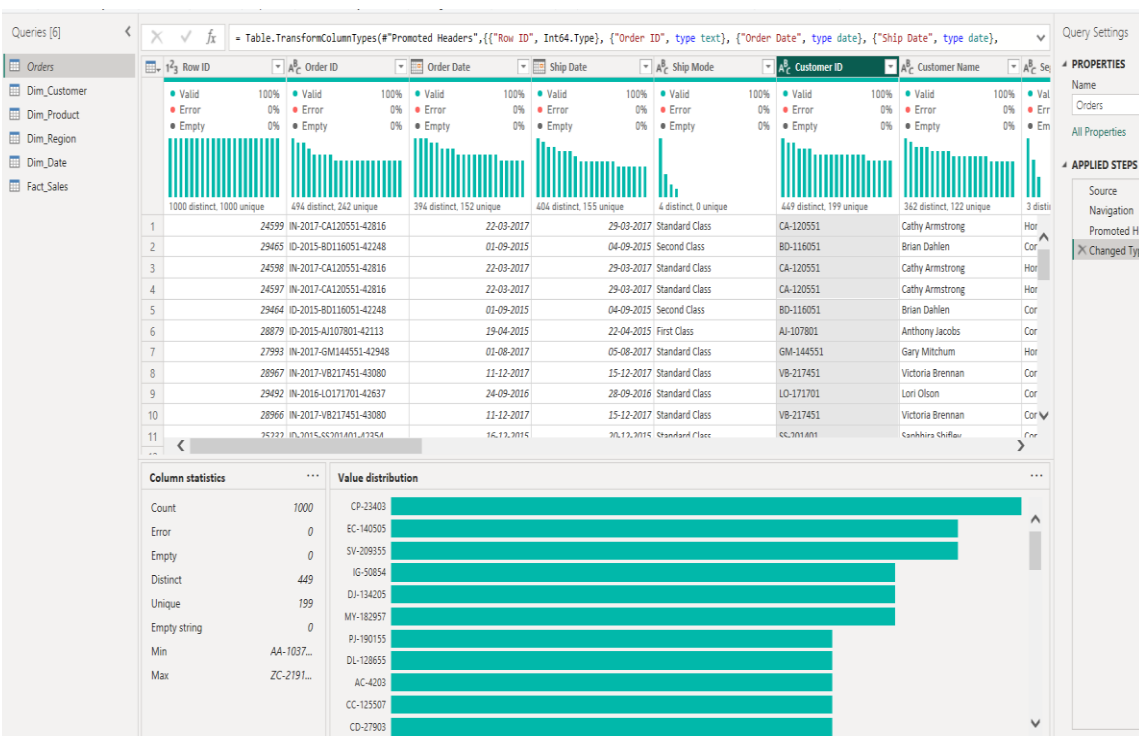
Step 21: Validate the Star Schema.
Use Customer, Product, Region and Date fields to filter or group Fact_Sales measures. Verify that dimension selections correctly change the dashboard values.

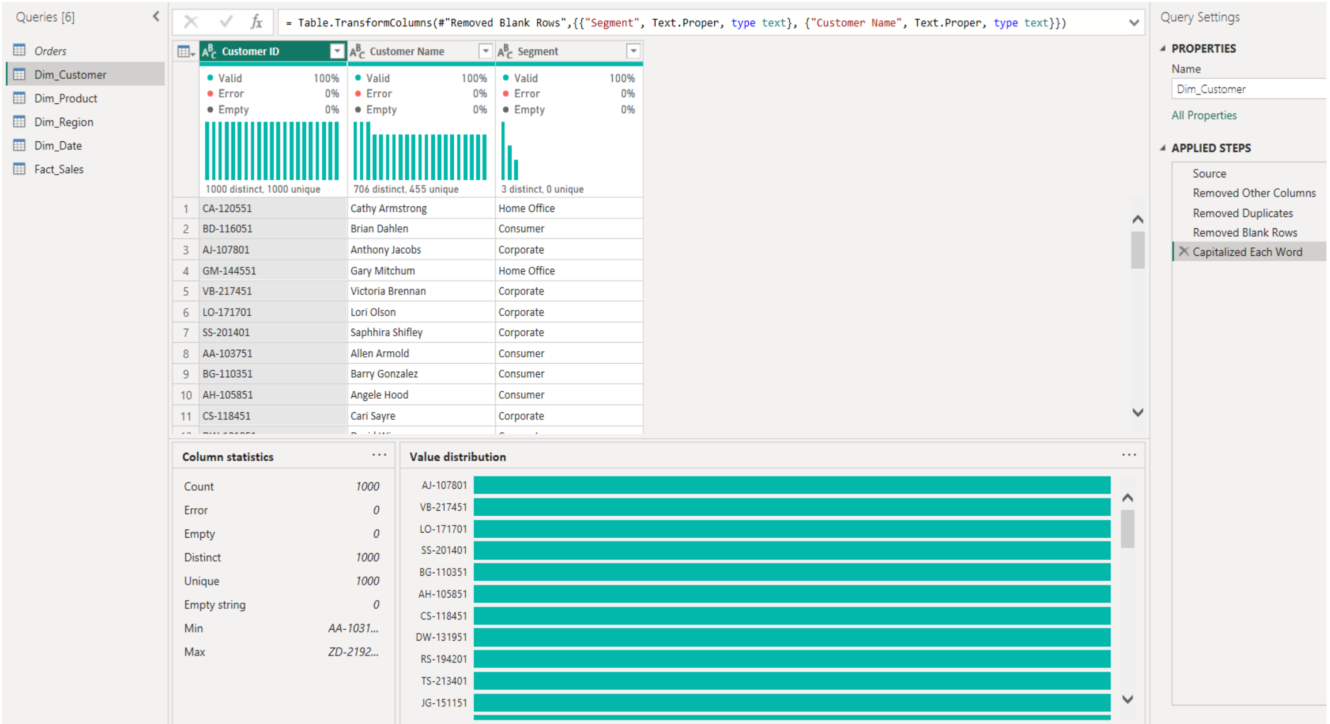
7. RESULT

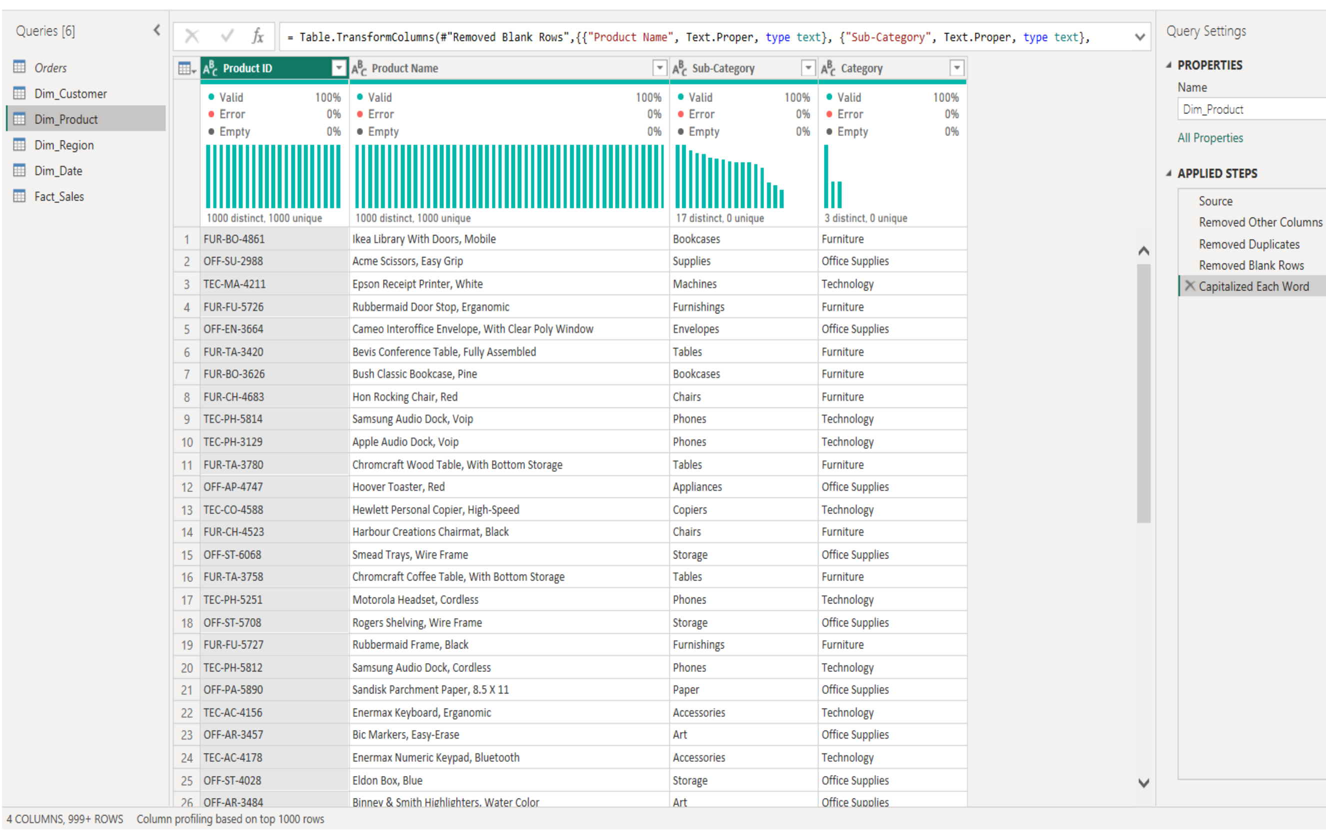
A dimensional data model was successfully designed and implemented in Microsoft Power BI using the Global Superstore dataset. The transactional data was organized into a central Fact_Sales table, while descriptive information was separated into Dim_Customer, Dim_Product, Dim_Region and Dim_Date. Relationships were established to form a Star Schema, DAX measures were created, and Power BI visualizations were used to validate the model for analytical reporting.

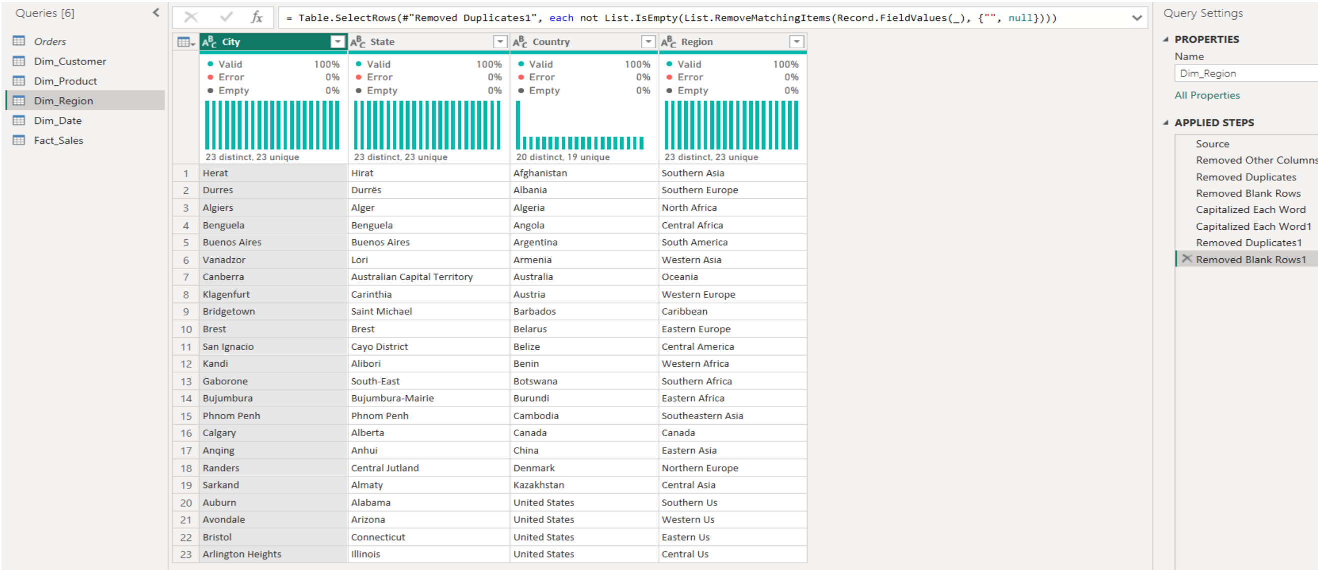


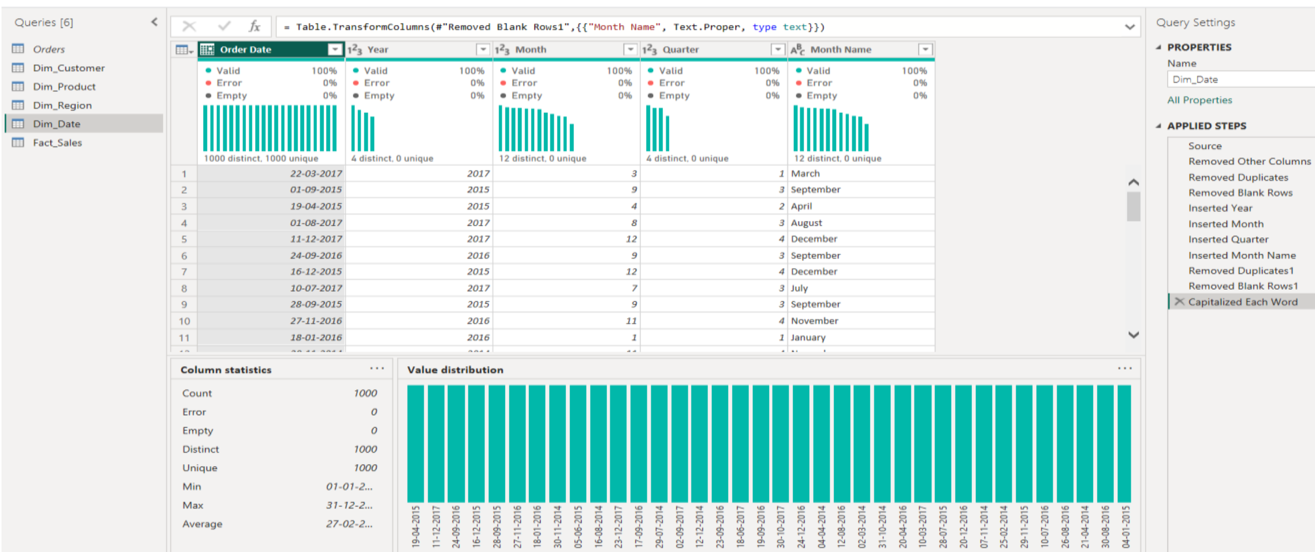

Dim_Region
City
Country
Region
State
Fact_Sales
Customer ID
Discount
Order Date
Order ID
Product ID
Profit
Quantity
Region
Dim_Customer
Customer ID
Customer Name
Segment
Dim_Date
Month
Month Name
Order Date
Quarter
Year
Dim_Product
Category
Product ID
Product Name
Sub-Category
1
*

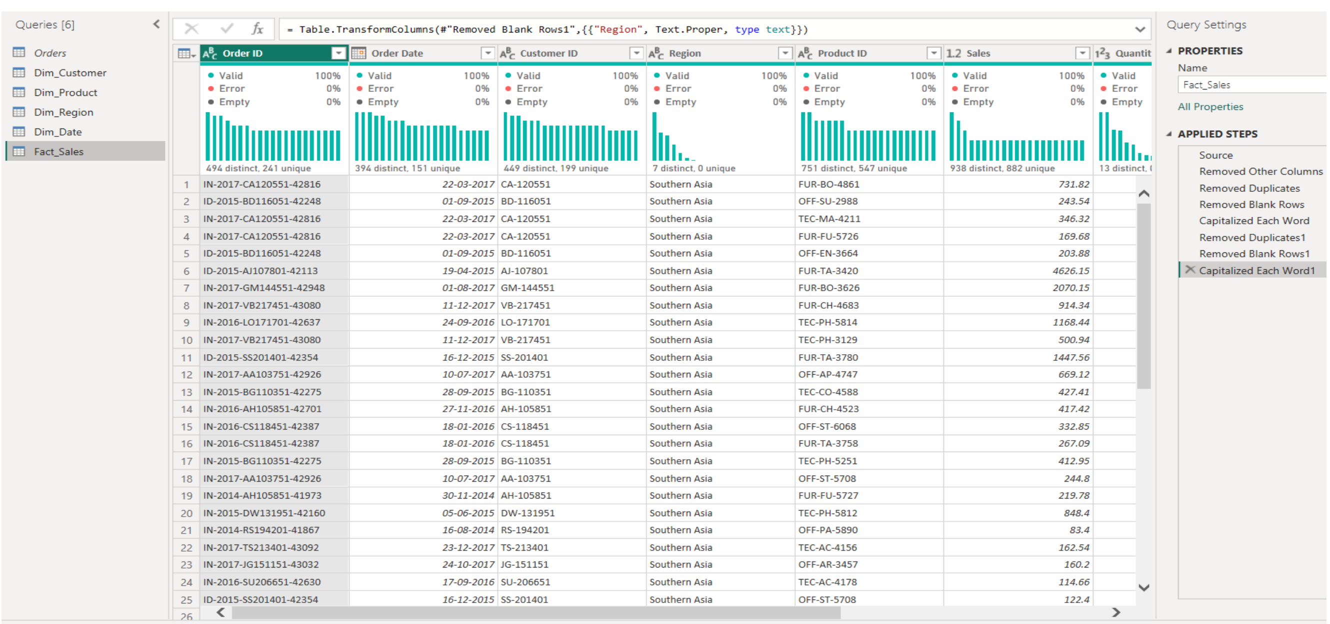

Queries [6]
Orders
Dim_Customer
Dim_Product
Dim_Region
Dim_Date
Fact_Sales
= Table.TransformColumnTypes(#"Promoted Headers",{{"Row ID", Int64.Type}, {"Order ID", type text}, {"Order Date", type date}, {"Ship Date", type date},
Row ID
Order ID
Order Date
Ship Date
Ship Mode
Customer ID
Customer Name
Valid
Error
Empty
100%
0%
1000 distinct, 1000 unique
494 distinct, 242 unique
394 distinct, 152 unique
404 distinct, 155 unique
4 distinct, 0 unique
449 distinct, 199 unique
362 distinct, 122 unique
1 24599 IN-2017-CA120551-42816 22-03-2017 29-03-2017 Standard Class CA-120551 Cathy Armstrong
2 29465 ID-2015-BD116051-42248 01-09-2015 04-09-2015 Second Class BD-116051 Brian Dahlen
3 24598 IN-2017-CA120551-42816 22-03-2017 29-03-2017 Standard Class CA-120551 Cathy Armstrong
4 24597 IN-2017-CA120551-42816 22-03-2017 29-03-2017 Standard Class CA-120551 Cathy Armstrong
5 29464 ID-2015-BD116051-42248 01-09-2015 04-09-2015 Second Class BD-116051 Brian Dahlen
6 28879 ID-2015-AJ107801-42113 19-04-2015 22-04-2015 First Class AJ-107801 Anthony Jacobs
7 27993 IN-2017-GM144551-42948 01-08-2017 05-08-2017 Standard Class GM-144551 Gary Mitchum
8 28967 IN-2017-VB217451-43080 11-12-2017 15-12-2017 Standard Class VB-217451 Victoria Brennan
9 29492 IN-2016-LO171701-42637 24-09-2016 28-09-2016 Standard Class LO-171701 Lori Olson
10 28966 IN-2017-VB217451-43080 11-12-2017 15-12-2017 Standard Class VB-217451 Victoria Brennan
Query Settings
PROPERTIES
Name
Orders
All Properties
APPLIED STEPS
Source
Navigation
Column statistics
Count 1000
Error 0
Empty 0
Distinct 449
Unique 199
Empty string 0
Min AA-1037...
Max ZC-2191...
Value distribution
CP-23403
EC-140505
SV-209355
IG-50854
DJ-134205
MY-182957
PJ-190155
DL-128655
AC-4203
CC-125507
CD-27903


Queries [6]
Orders
Dim_Customer
Dim_Product
Dim_Region
Dim_Date
Fact_Sales
= Table.TransformColumns(#"Removed Blank Rows",{{"Segment", Text.Proper, type text}, {"Customer Name", Text.Proper, type text}})
Customer ID
Customer Name
Segment
Valid 100%
Error 0%
Empty 0%
1000 distinct, 1000 unique
706 distinct, 455 unique
3 distinct, 0 unique
1 CA-120551 Cathy Armstrong Home Office
2 BD-116051 Brian Dahlen Consumer
3 AJ-107801 Anthony Jacobs Corporate
4 GM-144551 Gary Mitchum Home Office
5 VB-217451 Victoria Brennan Corporate
6 LO-171701 Lori Olson Corporate
7 SS-201401 Saphhira Shifley Corporate
8 AA-103751 Allen Armold Consumer
9 BG-110351 Barry Gonzalez Consumer
10 AH-105851 Angele Hood Consumer
11 CS-118451 Cari Sayre Corporate
Column statistics
Count 1000
Error 0
Empty 0
Distinct 1000
Unique 1000
Empty string 0
Min AA-1031...
Max ZD-2192...
Value distribution
AJ-107801
VB-217451
LO-171701
SS-201401
BG-110351
AH-105851
CS-118451
DW-131951
RS-194201
TS-213401
JG-151151
Query Settings
PROPERTIES
Name
Dim_Customer
All Properties
APPLIED STEPS
Source
Removed Other Columns
Removed Duplicates
Removed Blank Rows
Capitalized Each Word


Queries [6]
Orders
Dim_Customer
Dim_Product
Dim_Region
Dim_Date
Fact_Sales
= Table.TransformColumns(#"Removed Blank Rows",{{"Product Name", Text.Proper, type text}, {"Sub-Category", Text.Proper, type text},
Product ID
Product Name
Sub-Category
Category
Valid 100%
Error 0%
Empty 0%
1000 distinct, 1000 unique
1000 distinct, 1000 unique
17 distinct, 0 unique
3 distinct, 0 unique
1 FUR-BO-4861 Ikea Library With Doors, Mobile Bookcases Furniture
2 OFF-SU-2988 Acme Scissors, Easy Grip Supplies Office Supplies
3 TEC-MA-4211 Epson Receipt Printer, White Machines Technology
4 FUR-FU-5726 Rubbermaid Door Stop, Erganomic Furnishings Furniture
5 OFF-EN-3664 Cameo Interoffice Envelope, With Clear Poly Window Envelopes Office Supplies
6 FUR-TA-3420 Bevis Conference Table, Fully Assembled Tables Furniture
7 FUR-BO-3626 Bush Classic Bookcase, Pine Bookcases Furniture
8 FUR-CH-4683 Hon Rocking Chair, Red Chairs Furniture
9 TEC-PH-5814 Samsung Audio Dock, Voip Phones Technology
10 TEC-PH-3129 Apple Audio Dock, Voip Phones Technology
11 FUR-TA-3780 Chromcraft Wood Table, With Bottom Storage Tables Furniture
12 OFF-AP-4747 Hoover Toaster, Red Appliances Office Supplies
13 TEC-CO-4588 Hewlett Personal Copier, High-Speed Copiers Technology
14 FUR-CH-4523 Harbour Creations Chairmat, Black Chairs Furniture
15 OFF-ST-6068 Smead Trays, Wire Frame Storage Office Supplies
16 FUR-TA-3758 Chromcraft Coffee Table, With Bottom Storage Tables Furniture
17 TEC-PH-5251 Motorola Headset, Cordless Phones Technology
18 OFF-ST-5708 Rogers Shelving, Wire Frame Storage Office Supplies
19 FUR-FU-5727 Rubbermaid Frame, Black Furnishings Furniture
20 TEC-PH-5812 Samsung Audio Dock, Cordless Phones Technology
21 OFF-PA-5890 Sandisk Parchment Paper, 8.5 X 11 Paper Office Supplies
22 TEC-AC-4156 Enermax Keyboard, Erganomic Accessories Technology
23 OFF-AR-3457 Bic Markers, Easy-Erase Art Office Supplies
24 TEC-AC-4178 Enermax Numeric Keypad, Bluetooth Accessories Technology
25 OFF-ST-4028 Eldon Box, Blue Storage Office Supplies
26 OFF-AR-3484 Binney & Smith Highlighters, Water Color Art Office Supplies
Query Settings
PROPERTIES
Name
Dim_Product
All Properties
APPLIED STEPS
Source
Removed Other Columns
Removed Duplicates
Removed Blank Rows
Capitalized Each Word
4 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows


Queries [6]
Orders
Dim_Customer
Dim_Product
Dim_Region
Dim_Date
Fact_Sales
= Table.SelectRows(#"Removed Duplicates1", each not List.IsEmpty(List.RemoveMatchingItems(Record.FieldValues(_), {"", null})))
City
State
Country
Region
Valid 100%
Error 0%
Empty 0%
23 distinct, 23 unique
23 distinct, 23 unique
20 distinct, 19 unique
23 distinct, 23 unique
1 Herat Hirat Afghanistan Southern Asia
2 Durres Durrës Albania Southern Europe
3 Algiers Alger Algeria North Africa
4 Benguela Benguela Angola Central Africa
5 Buenos Aires Buenos Aires Argentina South America
6 Vanadzor Lori Armenia Western Asia
7 Canberra Australian Capital Territory Australia Oceania
8 Klagenfurt Carinthia Austria Western Europe
9 Bridgetown Saint Michael Barbados Caribbean
10 Brest Brest Belarus Eastern Europe
11 San Ignacio Cayo District Belize Central America
12 Kandi Alibori Benin Western Africa
13 Gaborone South-East Botswana Southern Africa
14 Bujumbura Bujumbura-Mairie Burundi Eastern Africa
15 Phnom Penh Phnom Penh Cambodia Southeastern Asia
16 Calgary Alberta Canada Canada
17 Anqing Anhui China Eastern Asia
18 Randers Central Jutland Denmark Northern Europe
19 Sarkand Almaty Kazakhstan Central Asia
20 Auburn Alabama United States Southern Us
21 Avondale Arizona United States Western Us
22 Bristol Connecticut United States Eastern Us
23 Arlington Heights Illinois United States Central Us
Query Settings
PROPERTIES
Name
Dim_Region
All Properties
APPLIED STEPS
Source
Removed Other Columns
Removed Duplicates
Removed Blank Rows
Capitalized Each Word
Capitalized Each Word1
Removed Duplicates1
Removed Blank Rows1


Queries [6]
Orders
Dim_Customer
Dim_Product
Dim_Region
Dim_Date
Fact_Sales
= Table.TransformColumns(#"Removed Blank Rows1",{{"Month Name", Text.Proper, type text}})
Order Date
Year
Month
Quarter
Month Name
Valid 100%
Error 0%
Empty 0%
1000 distinct, 1000 unique
4 distinct, 0 unique
12 distinct, 0 unique
4 distinct, 0 unique
12 distinct, 0 unique
1 22-03-2017 2017 3 1 March
2 01-09-2015 2015 9 3 September
3 19-04-2015 2015 4 2 April
4 01-08-2017 2017 8 3 August
5 11-12-2017 2017 12 4 December
6 24-09-2016 2016 9 3 September
7 16-12-2015 2015 12 4 December
8 10-07-2017 2017 7 3 July
9 28-09-2015 2015 9 3 September
10 27-11-2016 2016 11 4 November
11 18-01-2016 2016 1 1 January
Column statistics
Count 1000
Error 0
Empty 0
Distinct 1000
Unique 1000
Min 01-01-2...
Max 31-12-2...
Average 27-02-2...
Value distribution
Query Settings
PROPERTIES
Name
Dim_Date
All Properties
APPLIED STEPS
Source
Removed Other Columns
Removed Duplicates
Removed Blank Rows
Inserted Year
Inserted Month
Inserted Quarter
Inserted Month Name
Removed Duplicates1
Removed Blank Rows1
Capitalized Each Word


Queries [6]
Orders
Dim_Customer
Dim_Product
Dim_Region
Dim_Date
Fact_Sales
= Table.TransformColumns(#"Removed Blank Rows1",{{"Region", Text.Proper, type text}})
Order ID
Order Date
Customer ID
Region
Product ID
Sales
Quantit
Valid 100%
Error 0%
Empty 0%
494 distinct, 241 unique
394 distinct, 151 unique
449 distinct, 199 unique
7 distinct, 0 unique
751 distinct, 547 unique
938 distinct, 882 unique
13 distinct
1 IN-2017-CA120551-42816 22-03-2017 CA-120551 Southern Asia FUR-BO-4861 731.82
2 ID-2015-BD116051-42248 01-09-2015 BD-116051 Southern Asia OFF-SU-2988 243.54
3 IN-2017-CA120551-42816 22-03-2017 CA-120551 Southern Asia TEC-MA-4211 346.32
4 IN-2017-CA120551-42816 22-03-2017 CA-120551 Southern Asia FUR-FU-5726 169.68
5 ID-2015-BD116051-42248 01-09-2015 BD-116051 Southern Asia OFF-EN-3664 203.88
6 ID-2015-AJ107801-42113 19-04-2015 AJ-107801 Southern Asia FUR-TA-3420 4626.15
7 IN-2017-GM144551-42948 01-08-2017 GM-144551 Southern Asia FUR-BO-3626 2070.15
8 IN-2017-VB217451-43080 11-12-2017 VB-217451 Southern Asia FUR-CH-4683 914.34
9 IN-2016-LO171701-42637 24-09-2016 LO-171701 Southern Asia TEC-PH-5814 1168.44
10 IN-2017-VB217451-43080 11-12-2017 VB-217451 Southern Asia TEC-PH-3129 500.94
11 ID-2015-SS201401-42354 16-12-2015 SS-201401 Southern Asia FUR-TA-3780 1447.56
12 IN-2017-AA103751-42926 10-07-2017 AA-103751 Southern Asia OFF-AP-4747 669.12
13 IN-2015-BG110351-42275 28-09-2015 BG-110351 Southern Asia TEC-CO-4588 427.41
14 IN-2016-AH105851-42701 27-11-2016 AH-105851 Southern Asia FUR-CH-4523 417.42
15 IN-2016-CS118451-42387 18-01-2016 CS-118451 Southern Asia OFF-ST-6068 332.85
16 IN-2016-CS118451-42387 18-01-2016 CS-118451 Southern Asia FUR-TA-3758 267.09
17 IN-2015-BG110351-42275 28-09-2015 BG-110351 Southern Asia TEC-PH-5251 412.95
18 IN-2017-AA103751-42926 10-07-2017 AA-103751 Southern Asia OFF-ST-5708 244.8
19 IN-2014-AH105851-41973 30-11-2014 AH-105851 Southern Asia FUR-FU-5727 219.78
20 IN-2015-DW131951-42160 05-06-2015 DW-131951 Southern Asia TEC-PH-5812 848.4
21 IN-2014-RS194201-41867 16-08-2014 RS-194201 Southern Asia OFF-PA-5890 83.4
22 IN-2017-TS213401-43092 23-12-2017 TS-213401 Southern Asia TEC-AC-4156 162.54
23 IN-2017-JG151151-43032 24-10-2017 JG-151151 Southern Asia OFF-AR-3457 160.2
24 IN-2016-SU206651-42630 17-09-2016 SU-206651 Southern Asia TEC-AC-4178 114.66
25 ID-2015-SS201401-42354 16-12-2015 SS-201401 Southern Asia OFF-ST-5708 122.4
Query Settings
PROPERTIES
Name
Fact_Sales
All Properties
APPLIED STEPS
Source
Removed Other Columns
Removed Duplicates
Removed Blank Rows
Capitalized Each Word
Removed Duplicates1
Removed Blank Rows1
Capitalized Each Word1
9 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

7.1 OBSERVATIONS

- Fact_Sales stores transaction-level measures and relationship keys.
- Dimension tables provide descriptive context for analysing the facts.
- The Star Schema provides a clear and structured BI data model.
- DAX measures respond dynamically to filters from the dimensions.
- The model supports analysis by customer, product, region and time.

8. LEARNING OUTCOMES

- Understood Dimensional Modeling and Star Schema.
- Learned the difference between Fact and Dimension Tables.
- Gained practical experience creating dimensional tables in Power Query.
- Learned to create and validate relationships between fact and dimensions.
- Learned to create DAX measures for analytical reporting.
- Understood Total Sales, Total Profit, Profit Margin and Average Order Value.
- Learned to validate a Star Schema through Power BI visualizations.
- Gained practical experience building a structured BI model for reporting and decision-making.

9. CONCLUSION

The experiment successfully demonstrated dimensional modeling in Microsoft Power BI. By separating transaction-level data from descriptive attributes and connecting the resulting tables through a Star Schema, the Global Superstore dataset was organized into a structured analytical model. Power Query supported data preparation, Model View supported relationship management, and DAX supported dynamic business calculations.